

$$\int \frac{dx}{\sin 2x + 2\sin x} = \int \frac{dx}{2\sin x(\cos x + 1)}$$

$$= \int \frac{\sin x dx}{2(1 - \cos^2 x)(1 + \cos x)} \quad \frac{1/4}{1-u} + \frac{1/4}{1+u} + \frac{1/2}{(1+u)^2}$$

$$\underline{\underline{\cos x = u}} - \frac{1}{2} \int \frac{du}{(1+u)(1-u^2)}$$

$$= \frac{1}{8} (\ln |1-u| - \ln |1+u| + \frac{2}{1+u}) + C$$

$$= \frac{1}{8} \left[\ln \frac{1 - \cos x}{1 + \cos x} + \frac{2}{1 + \cos x} \right] + C$$

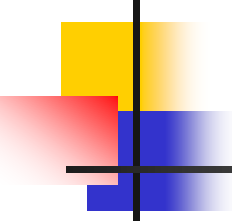
5. 计算不定积分: $\int \frac{dx}{x(x^n + a)}, \int \frac{x \ln(x + \sqrt{1+x^2})}{(1-x^2)^2} dx;$

解: $\int \frac{dx}{x(x^n + a)} = \frac{1}{a} \int \frac{x^n + a - x^n}{x(x^n + a)} dx$

$$= \frac{1}{a} \ln |x| - \frac{1}{an} \ln |x^n + a| + C$$

$$\int \frac{x \ln(x + \sqrt{1+x^2})}{(1-x^2)^2} dx = \frac{1}{2} \int \ln(x + \sqrt{1+x^2}) d \frac{1}{1-x^2}$$

$$= \frac{1}{2} \cdot \frac{1}{1-x^2} \ln(x + \sqrt{1+x^2}) - \int \frac{1}{1-x^2} \cdot (\ln(x + \sqrt{1+x^2}))' dx$$


$$= \frac{1}{2} \cdot \frac{1}{1-x^2} \ln(x + \sqrt{1+x^2}) - \frac{1}{2} \int \frac{1}{1-x^2} \cdot \frac{1}{\sqrt{1+x^2}} dx$$

$$\text{令 } x = \tan t, dx = \sec^2 t dt$$

$$\int \frac{dx}{(1-x^2)\sqrt{1+x^2}} = \int \frac{\sec^2 t dt}{(1-\tan^2 t)\sec t}$$

$$= \int \frac{\cos t dt}{\cos^2 t - \sin^2 t} = \int \frac{d \sin t}{1-2\sin^2 t} \cdot \frac{1}{2} \left(\frac{1}{1-\sqrt{2}\sin t} + \frac{1}{1+\sqrt{2}\sin t} \right)$$

$$= \frac{1}{2\sqrt{2}} \ln \frac{1+\sqrt{2}\sin t}{1-\sqrt{2}\sin t} + C$$

7. 设 $f(x)$ 在 $[a,b]$ 上连续, 且 $f(x) > 0$, 又


$$F(x) = \int_a^x f(t)dt + \int_b^x \frac{1}{f(t)}dt, \text{ 试证: } F'(x) \geq 2, \text{ 且}$$

$F(x) = 0$ 在 $[a,b]$ 内仅有一个实根;

证明: $F'(x) = f(x) + \frac{1}{f(x)} \geq 2\sqrt{f(x) \cdot \frac{1}{f(x)}} = 2$

$$F(a) = \int_b^a \frac{1}{f(x)}dx = \frac{1}{f(\xi)}(a-b) < 0, \quad \xi \in [a,b]$$

$$F(b) = \int_a^b f(x)dx = f(\eta)(b-a) > 0, \quad \eta \in [a,b]$$

$F(x)$ 是连续函数, $F'(x) \geq 2 > 0$, 单调,

故仅有一根。

9. 设 $f(x)$ 有二阶连续导数, 且 $f'(0) = 0, \lim_{x \rightarrow 0} \frac{f''(x)}{|x|} = 1$,

则 $f(0)$ 是 $f(x)$ 的极小值;

解: 由 $\lim_{x \rightarrow 0} \frac{f''(x)}{|x|} = 1$, 存在 $x = 0$ 的某邻域, 在此邻域内有

$\frac{f''(x)}{|x|} \geq 0$, 即 $f''(x) \geq 0$, 又由 Taylor 公式

$$f(x) = f(0) + f'(0)x + \frac{f''(\xi)}{2}x^2 \quad \xi \text{ 介于 } 0 \text{ 和 } x \text{ 之间}$$

$$\text{从而 } f(x) = f(0) + \frac{f''(\xi)}{2}x^2 \geq f(0)$$

可见 $f_{\min} = f(0)$.

12. 设 $f(x)$ 连续, $\varphi(x) = \int_0^1 f(xt)dt$, 且 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = A$ (A 为

常数) 求 $\varphi'(x)$, 并讨论 $\varphi'(x)$ 在 $x=0$ 处的连续性;

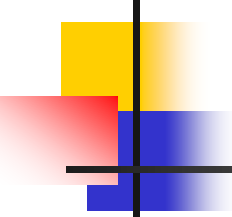
解: $\varphi(x) = \frac{1}{x} \int_0^x f(t)dt$, 由 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = A$

$$\Rightarrow f(0) = 0, f'(0) = A, \text{ 且有 } \varphi(0) = 0.$$

$$\therefore \varphi'(x) = \frac{xf(x) - \int_0^x f(u)du}{x^2} \quad (x \neq 0)$$

由导数定义, 有

$$\varphi'(0) = \lim_{x \rightarrow 0} \frac{\int_0^x f(u)du}{x^2} = \lim_{x \rightarrow 0} \frac{f(x)}{2x} = \frac{A}{2}$$



而 $\lim_{x \rightarrow 0} \varphi'(x) = \lim_{x \rightarrow 0} \frac{xf(x) - \int_0^x f(u)du}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{f(x)}{x} - \lim_{x \rightarrow 0} \frac{\int_0^x f(u)du}{x^2}$$

$$= A - \frac{A}{2} = \varphi'(0)$$

可见 $\varphi'(x)$ 在 $x = 0$ 处连续.

14. 设 $f(x) = \begin{cases} \frac{1 - \cos x}{\sqrt{x}}, & x > 0 \\ x^2 g(x), & x \leq 0. \end{cases}$, 其中 $g(x)$ 是有界函数,

则 $f(x)$ 在 $x = 0$ 处连续, 可导;

解: $f'_+(0) = \lim_{x \rightarrow 0+} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0+} \frac{1 - \cos x}{x^{\frac{3}{2}}} = 0$

$$f'_-(0) = \lim_{x \rightarrow 0-} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0-} \frac{x^2 g(x)}{x} = 0$$

$\therefore f(x)$ 在 $x = 0$ 处可导, 则连续.

15. 证明：当 $x > 0$ 时, $(x^2 - 1)\ln x \geq (x - 1)^2$;

证明：令 $f(x) = (x^2 - 1)\ln x - (x - 1)^2$, $f(1) = 0$

$$f'(x) = 2x\ln x - x + 2 - \frac{1}{x}, \quad f'(1) = 0$$

$$f''(x) = 2\ln x + 3 + \frac{1}{x^2}, \quad f''(1) = 2 > 0$$

$$f'''(x) = \frac{2(x^2 - 1)}{x^3}$$

$$0 < x < 1 \text{ 时, } f'''(x) < 0$$

$$1 < x < +\infty \text{ 时, } f'''(x) > 0$$

$$\Rightarrow 0 < x < +\infty, f''(x) \geq f''(1) = 2 > 0$$

又由 $f'(1) = 0$ 及 $f'(x)$ 是单调增函数知


$$0 < x < 1 \text{ 时, } f'(x) < 0$$

$$1 < x < +\infty \text{ 时, } f'(x) > 0$$

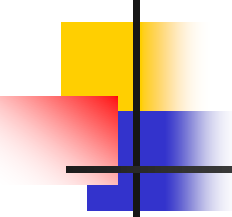
$$\Rightarrow f(x) > f(1) = 0 \quad (0 < x < +\infty)$$

即 $x > 0$ 时, $(x^2 - 1)\ln x > (x - 1)^2$.

另, 适当变形, 当 $x > 0$ 时, $(x^2 - 1)\ln x \geq (x - 1)^2$ 等价于

$$0 < x < 1 \text{ 时, } \ln x \leq \frac{x - 1}{x + 1}$$

$$1 < x < +\infty \text{ 时, } \ln x \geq \frac{x - 1}{x + 1}$$


$$\text{令 } f(x) = \ln x - \frac{x-1}{x+1}$$

$$\text{则 } f'(x) = \frac{1}{x} - \frac{2}{(x+1)^2} = \frac{x^2+1}{x(x+1)^2} > 0$$

又 $f(1) = 0$, 可见有

$0 < x < 1$ 时, $f(x) < 0$, $1 < x < +\infty$ 时, $f(x) > 0$.

从而 $(x^2 - 1)f(x) = (x^2 - 1)\ln x - (x - 1)^2 \geq 0$

即 $x > 0$ 时, $(x^2 - 1)\ln x \geq (x - 1)^2$

法3 列表判别:

$$f(x) = (x^2 - 1)\ln x - (x - 1)^2, \quad f(1) = 0$$

$$f'(x) = 2x \ln x - \frac{1}{x} + 2, \quad f'(1) = 0$$

$$f''(x) = 2 \ln x + \frac{1}{x^2} + 1, \quad f''(1) = 2 > 0$$

$$f'''(x) = \frac{2(x^2 - 1)}{x^3}$$

x	$(0, 1)$	1	$(1, +\infty)$
$f'''(x)$	$-$	0	$+$
$f''(x)$	 $+$	2	 $+$
$f'(x)$	 $-$	0	 $+$
$f(x)$	 $+$	0	 $+$

故当 $x > 0$ 时 $f(x) \geq 0$, 即 $(x^2 - 1)\ln x \geq (x - 1)^2$.

$$f(x) = (x^2 - 1)\ln x - (x - 1)^2, \quad f(1) = 0$$

$$f'(x) = 2x\ln x - \frac{1}{x} + 2 \quad f'(1) = 0$$

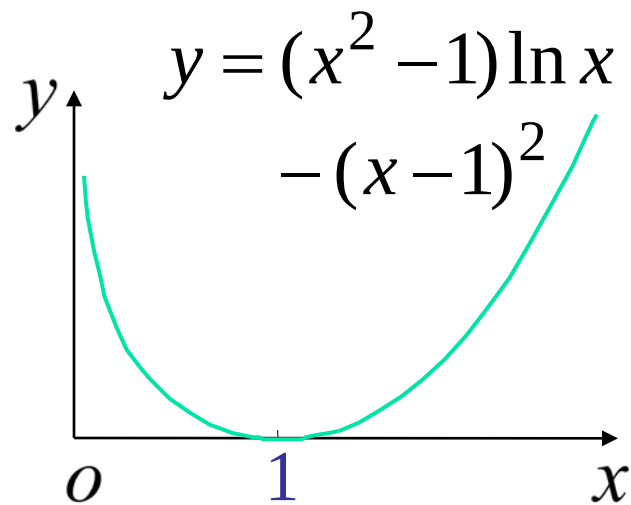
$$f''(x) = 2\ln x + \frac{1}{x^2} + 1, \quad f''(1) = 2 > 0$$

法4 利用极值第二判别法.

易知 $x = 1$ 是 $f'(x) = 0$ 的唯一根,
且 $f''(1) > 0, \therefore x = 1$ 为 $f(x)$ 的唯一
极小点, 故 $f(1) = 0$ 也是最小值,

因此当 $x > 0$ 时 $f(x) \geq 0$, 即

$$(x^2 - 1)\ln x \geq (x - 1)^2$$



16. 设 $f(x)$ $(-\infty, +\infty)$ 内连续, 并且对任意的 x

$$f(x) + 4 \int_0^x t f(x-t) dt = \frac{1}{3} x^3$$

1) 求证: 对任意的 x , $\int_0^x t f(x-t) dt = x \int_0^x f(t) dt - \int_0^x t f(t) dt$;

2) 试求出 $f(x)$ 的表达式.

证明: 做变换 $u = x - t$, 则

$$\begin{aligned} \int_0^x t f(x-t) dt &= \int_x^0 (x-u) f(u) (-1) du \\ &= \int_0^x (x-u) f(u) du = x \int_0^x f(t) dt - \int_0^x t f(t) dt \quad (1) \end{aligned}$$

将 (1) 的结果代入原方程, 等式两边关于 x 求导, 得

$$f'(x) + 4 \int_0^x f(t) dt = x^2$$



由原方程及上式可知 $f(0) = 0, f'(0) = 0$

且 $f''(x) + 4f(x) = 2x$

上述二阶微分方程的通解为：

$$f(x) = C_1 \cos 2x + C_2 \sin 2x + \frac{x}{2}$$

将初值条件代入，得

$$f(x) = -\frac{1}{4} \sin 2x + \frac{x}{2}$$

17. 设函数 $f(x)$ 在 $(-\infty, +\infty)$ 内具有二阶导数, 且 $y' \neq 0$,

$x = x(y)$ 是 $y = y(x)$ 的反函数

1) 试将 $x = x(y)$ 所满足的微分方程

$$\frac{d^2 x}{dy^2} + (y + \sin x) \left(\frac{dx}{dy} \right)^3 = 0 \text{ 变换为 } y = y(x) \text{ 满足的方程;}$$

2) 求变换后的微分方程满足初始条件 $y(0) = 0, y'(0) = \frac{3}{2}$ 的解;

解: $\frac{dx}{dy} = \frac{1}{y'}$, 于是

$$\frac{d^2 x}{dy^2} = \frac{d}{dx} \left(\frac{1}{y'} \right) \cdot \frac{dx}{dy} = \frac{-y''}{y'^2} \cdot \frac{1}{y'} = -\frac{y''}{y'^3}$$

代入原方程得： $y'' - y = \sin x$

对应齐次方程通解为： $Y = C_1 e^x + C_2 e^{-x}$

设非齐次方程特解为： $y^* = A \cos x + B \sin x$

代入原方程得： $A = 0, B = -\frac{1}{2}$

$\therefore$ 原方程的通解为： $y = -\frac{1}{2} \sin x + C_1 e^x + C_2 e^{-x}$

由初始条件 $y(0) = 0, y'(0) = \frac{3}{2} \Rightarrow C_1 = 1, C_2 = -1$

$$y = e^x - e^{-x} - \frac{1}{2} \sin x.$$



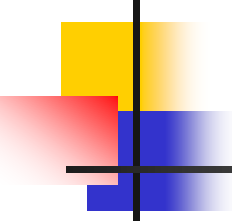
18. 设 $e < a < b < e^2$, 证明: $\ln^2 b - \ln^2 a > \frac{4}{e^2}(b - a)$;

证明: 对数函数 $y = \ln^2 x$ 在 $[a, b]$ 上应用 *Lagrange* 中值定理, 得

$$\ln^2 b - \ln^2 a = \frac{2\ln \xi}{\xi}(b - a), \quad a < \xi < b$$

设 $\varphi(t) = \frac{\ln t}{t}$, 则 $\varphi'(t) = \frac{1 - \ln t}{t^2}$,

$t > e$ 时, $\varphi'(t) < 0$, 所以 $\varphi(t)$ 单调减少, 从而
 $\varphi(\xi) > \varphi(e^2) (\because e < a < \xi < b < e^2)$


$$\frac{\ln \xi}{\xi} > \frac{\ln e^2}{e^2} = \frac{2}{e^2}$$

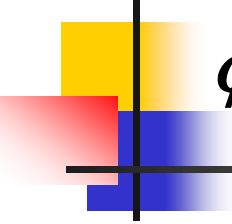
$$\text{故 } \ln^2 b - \ln^2 a > \frac{4}{e^2}(b - a).$$

另法： 设 $\varphi(x) = \ln^2 x - \frac{4}{e^2}x$ ， 则

$$\varphi'(x) = 2 \cdot \frac{\ln x}{x} - \frac{4}{e^2}, \quad \varphi''(x) = 2 \cdot \frac{1 - \ln x}{x^2},$$

$\therefore x > e$ 时， $\varphi''(x) < 0$ ， 故 $\varphi'(x)$ 单调减少，

从而当 $e < x < e^2$ 时，


$$\varphi'(x) > \varphi'(e^2) = \frac{4}{e^2} - \frac{4}{e^2} = 0,$$

即当 $e < x < e^2$ 时, $\varphi(x)$ 单调增加.

因此当 $e < x < e^2$ 时, $\varphi(b) > \varphi(a)$,

$$\text{即 } \ln^2 b - \frac{4}{e^2} b > \ln^2 a - \frac{4}{e^2} a$$

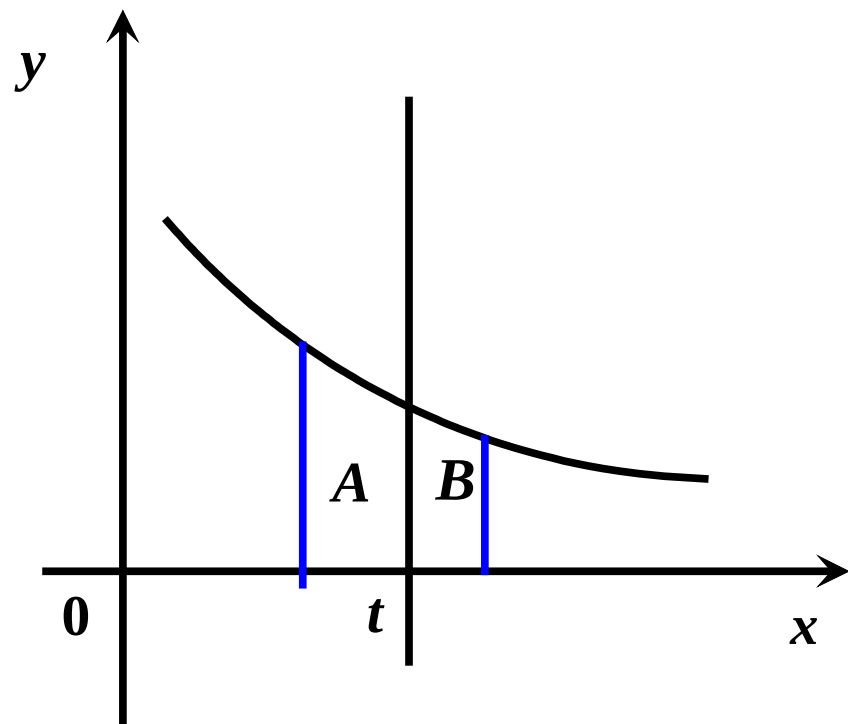
$$\text{故 } \ln^2 b - \ln^2 a > \frac{4}{e^2} (b - a).$$

23. 由 $y = \frac{1}{x^2}$, $y = 0$, $x = 1$, $x = 2$ 所围成的曲边梯形, 被直线 $x = t$ 分成 A, B 两部分, A, B 也分别代表两部分的面积, 将 A, B 分别绕 $x = t$ 旋转所成旋转体的体积分别记为 V_A, V_B , 求使 $V_A + V_B$ 取得最小值的 t 值.

解:
$$V_A = \int_1^t 2\pi(t-x) \frac{1}{x^2} dx$$

$$= 2\pi t \int_1^t \frac{dx}{x^2} - 2\pi \int_1^t \frac{1}{x} dx$$

$$= 2\pi(t-1) - 2\pi \int_1^t \frac{dx}{x}$$



$$V_B = \int_t^2 2\pi(x-t) \frac{dx}{x^2} = 2\pi \int_t^2 \frac{dx}{x} - 2\pi t \int_t^2 \frac{dx}{x^2}$$

$$= 2\pi \int_t^2 \frac{dx}{x} + 2\pi t \left(\frac{1}{2} - \frac{1}{t} \right) = 2\pi \int_t^2 \frac{dx}{x} + \pi(t-2)$$

$$V_A + V_B = \pi(t-2) + 2\pi \int_t^2 \frac{dx}{x} + 2\pi(t-1) - 2\pi \int_1^t \frac{dx}{x}$$

$$= 2\pi \left[\int_t^2 \frac{dx}{x} - \int_t^1 \frac{dx}{x} \right] + 3\pi t - 4\pi$$

$$(V_A + V_B)' = 2\pi \left(-\frac{2}{t} \right) + 3\pi = \frac{\pi(3t-4)}{t}$$

驻点为: $t = \frac{4}{3}$, $t < \frac{4}{3}$, $(V_A + V_B) < 0$, $t > \frac{4}{3}$, $(V_A + V_B) > 0$

$$\therefore V_{\min} = V\left(\frac{4}{3}\right) = 2\pi \ln \frac{9}{8}.$$

28. 已知 $f(x)$ 在 $x=0$ 处可导, 且对任意的实数 x, y , 满足方程 $f(x+y) = f(x) + f(y)$, 证明:

(1) $f(x)$ 在 $(-\infty, +\infty)$ 上连续;

(2) $f(x) = ax$, 其中 $a = f'(0)$ 。

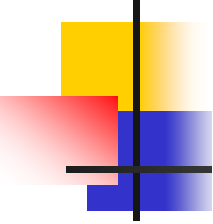
证明: (1) 由条件 $f(0) = f(0) + f(0)$, 得 $f(0) = 0$

$$\forall x \in (-\infty, +\infty), \Delta x \neq 0, \quad f(x + \Delta x) = f(x) + f(\Delta x),$$

$$\therefore \lim_{\Delta x \rightarrow 0} [f(x + \Delta x) - f(x)] = \lim_{\Delta x \rightarrow 0} f(\Delta x) = f(0) = 0$$

即 $f(x)$ 在 $(-\infty, +\infty)$ 连续。

(2) $\forall x \in (-\infty, +\infty)$, 由,


$$\begin{aligned}\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x) - f(0)}{\Delta x} = f'(0) = a\end{aligned}$$

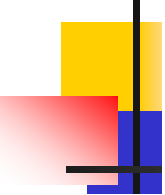
即 $\forall x \in (-\infty, +\infty)$, $f'(x) = f'(0) = a$

所以, 由 *Lagrange* 中值定理, $\forall x \in (-\infty, +\infty)$,

$$f(x) = f(x) - f(0) = f'(\xi)x = ax \quad \xi \text{ 介于 } 0, x \text{ 之间}$$

即 $f(x) = ax$, $\forall x \in (-\infty, +\infty)$,

33. 设 $c > 0$ ，任取 $0 < x_0 < \frac{1}{c}$ ，做迭代序列：


$$x_{n+1} = x_n(2 - cx_n), \quad n = 1, 2, \dots, \quad \text{求} \lim_{n \rightarrow \infty} x_n.$$

解： $cx_{n+1} = cx_n(2 - cx_n) = 1 - (1 - cx_n)^2$

由数学归纳法：由 $0 < cx_0 < 1 \Rightarrow 0 < cx_n < 1, \forall n \in N$

$$\Rightarrow \begin{cases} 0 < x_n < \frac{1}{c} \\ \frac{x_{n+1}}{x_n} = 2 - cx_n > 1 \Rightarrow x_{n+1} > x_n \end{cases}$$

由单调有界原理， $\{x_n\}$ 有极限，记为 A ，

等式两边取极限，得 $A = \frac{1}{c}$

答疑时间：1月7日9:00-11:30

答疑教师：闫志忠，张伟

答疑地点：文萃楼 F304

答疑时间：1月7日13:00-15:30

答疑教师：闫晓霞，张文娟

答疑地点：文萃楼 F304

答疑时间：1月8日9:00-11:30

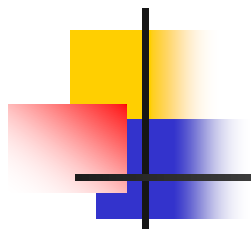
答疑教师：林甲富，孙兵

答疑地点：文萃楼 F304

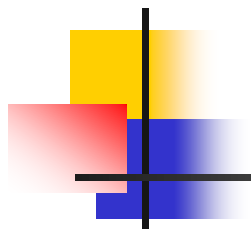
答疑时间：1月8日13:00-15:30

答疑教师：万捷，李国鹏

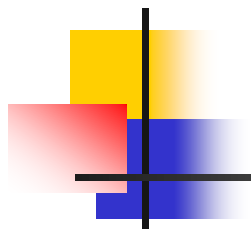
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用两种方法证明: 设函数 $f(x)$ 在 $[0, \pi]$ 上连续, 且 $\int_0^{\pi} f(x) dx = 0$,

$\int_0^{\pi} f(x) \cos x dx = 0$, 试证: 在 $(0, \pi)$ 内存在至少两

个不同的点 ξ_1, ξ_2 , 使 $f(\xi_1) = f(\xi_2) = 0$.

19. 设 $f(x)$ 在 $(-1,1)$ 内具有二阶连续导数,

且 $f''(x) \neq 0$, 试证:

1) 对于 $(-1,1)$ 内任一 $x \neq 0$, 存在唯一 $\theta(x) \in (0,1)$,

使 $f(x) = f(0) + xf'(\theta(x)x)$ 成立;

$$2) \lim_{x \rightarrow 0} \theta(x) = \frac{1}{2};$$

证明: 任给非零 $x \in (-1,1)$, 由 *Lagrange* 中值定理得

$$f(x) = f(0) + xf'(\theta(x)x) \quad (0 < \theta(x) < 1)$$

$\because f''(x)$ 在 $(-1,1)$ 内连续且 $f''(x) \neq 0$,

$\therefore f''(x)$ 在 $(-1,1)$ 内不变号, 不妨设 $f''(x) > 0$



则 $f'(x)$ 在 $(-1,1)$ 内严格单调增加, 故 $\theta(x)$ 唯一;

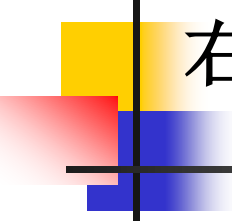
对于非零的 $x \in (-1,1)$, 由 *Langrange* 中值定理得:

$$f(x) = f(0) + xf'(\theta(x)x), (0 < \theta(x) < 1)$$

于是有:
$$\frac{f'(\theta(x)x) - f'(0)}{x} = \frac{f(x) - f(0) - f'(0)x}{x^2}$$

上式两端取极限, 得

$$\text{左端} = \lim_{x \rightarrow 0} \frac{f'(\theta(x)x) - f'(0)}{\theta(x)x} \cdot \theta(x) = f''(0) \lim_{x \rightarrow 0} \theta(x).$$


$$\text{右端} = \lim_{x \rightarrow 0} \frac{f'(x) - f'(0)}{2x} = \frac{1}{2} f''(0)$$

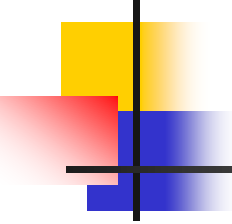
$$\Rightarrow \lim_{x \rightarrow 0} \theta(x) = \frac{1}{2}.$$

另法1 1) 同上

2) 由Taylor公式

$$f(x) = f(0) + f'(0)x + \frac{1}{2} f''(\xi)x^2, \xi \text{ 介于 } 0 \text{ 与 } x \text{ 之间}$$

$$\therefore x f'(\theta(x)x) = f(x) - f(0) = f'(0)x + \frac{1}{2} f''(\xi)x^2$$



从而 $\theta(x) \cdot \frac{f'(\theta(x)x) - f'(0)}{\theta(x)x} = \frac{1}{2} f''(\xi)$

由于 $\lim_{x \rightarrow 0} \frac{f'(\theta(x)x) - f'(0)}{\theta(x)x} = f''(0),$

$$\lim_{x \rightarrow 0} f''(\xi) = \lim_{\xi \rightarrow 0} f''(\xi) = f''(0)$$

故 $\lim_{x \rightarrow 0} \theta(x) = \frac{1}{2}.$



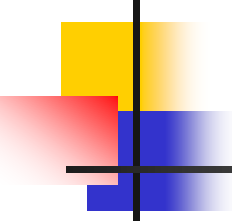
另法2 1) 同上

2) 因 $f''(x) \neq 0$, 故 $f'(x)$ 存在单值连续可导的反函数, 记为 $\varphi(x)$, 则有

$$\theta(x) \cdot x = \varphi\left(\frac{f(x) - f(0)}{x}\right), \quad \therefore \lim_{x \rightarrow 0} \varphi\left(\frac{f(x) - f(0)}{x}\right) = 0$$

$$\lim_{x \rightarrow 0} \theta(x) = \lim_{x \rightarrow 0} \frac{\varphi\left(\frac{f(x) - f(0)}{x}\right)}{x}$$

$$= \lim_{x \rightarrow 0} \varphi'\left(\frac{f(x) - f(0)}{x}\right) \cdot \frac{xf'(x) - f(x) + f(0)}{x^2}$$



$$= \varphi'(f'(0)) \lim_{x \rightarrow 0} \frac{xf''(x)}{2x} = \frac{1}{2} \varphi'(f'(0)) f''(0)$$

但因 $\varphi(f'(x)) = x$, 两边对 x 求导, 有

$$\varphi'(f'(x)) f''(x) = 1$$

以 $x = 0$ 代入, 于是有

$$\lim_{x \rightarrow 0} \theta(x) = \frac{1}{2}.$$